



CSE6324-SOFTWARE VERIFICATION & VALIDATION

Topic: Hospital Management System

Student Name	Student ID	Email
CHIN ZHEN HO	1221102540	1221102540@student.mmu.edu.my
PANG TIAN YOU	1231303394	1231303394@student.mmu.edu.my
LIM KAI SHEN	1211110602	1211110602@student.mmu.edu.my

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Test Plan

Hospital Management System (HMS)

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Prepared By: CHIN ZHEN HO, PANG TIAN YOU, LIM KAI SHEN	Group member	CHIN,PANG,LIM
Reviewed By: CHIN ZHEN HO	Leader	CHIN
Approved By: Ms. Venushini A/P Rajendran	Lecturer	

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1.0 Introduction

1.1 Purpose

The purpose of this document is to define the testing plan for the Hospital Management System (HMS). This test plan serves as a guide for the testing team to verify and validate that the application functions correctly and satisfies the specified requirements in the Software Requirements Specification (SRS). The plan covers both manual testing and automated testing activities to ensure that the major system features operate correctly under normal and abnormal conditions.

This document helps the team:

- Define the scope and objectives of the testing;
- Identify the items to be tested;
- Determine the test approach and testing strategy;
- Establish entry and exit criteria; and
- Ensure quality assurance activities are carried out in a structured and traceable manner.

1.2 Scope

This test plan applies to the Hospital Management System (HMS), which is a web-based application developed using PHP, MySQL, HTML, CSS, JavaScript, Bootstrap, and TCPDF. The system is designed to support hospital appointment handling and administrative operations, including patient, doctor, and admin functions.

Testing will cover the major functional modules of the system, including but not limited to:

- Patient registration, login, and logout
- Doctor login and doctor functions
- Admin login and admin functions

- Appointment booking and appointment cancellation
- Prescription creation and prescription viewing
- Bill PDF generation
- Viewing and searching of records
- Contact/query submission and query management

These activities will test both the front and back-end of the system with valid and invalid inputs to make sure it behaves correctly. Within the scope of the tests, we will also include some automated functional tests on critical scenarios for login, appointments, and doctor/admin, if applicable.

1.3 References

This document is prepared in reference to the following documents:

- i. ISO/IEC/IEEE 29148, Systems and software engineering – Life cycle processes – Requirements engineering
- ii. IEEE Std 1012-2016, IEEE Standard for System, Software and Hardware Verification and Validation
- iii. <https://github.com/kishan0725/Hospital-Management-System.git>

The following documents provide the test basis for this test plan:

- i. Hospital Management System (HMS) Software Requirements Specification (SRS), Version 1.0
- ii. HMS use case specifications and functional requirements
- iii. HMS source code and implemented interfaces

- iv. HMS database schema and related project documentation

2.0 Details of Test Plan

2.1 Test Items

The following documents and system components will serve as the foundation for defining the expected behavior and correct operation of the Hospital Management System (HMS):

- i. Hospital Management System (HMS) Software Requirements Specification (SRS), Version 1.0
- ii. Hospital Management System source code / [project repository](#)
- iii. Database schema file 'myhmsdb.sql'
- iv. Implemented user interfaces, workflows, and supporting project documentation
- v. Recorded browser-based test flows and reusable automated test scripts created using Katalon Recorder
- vi. Test step records, screenshots, and generated manual test case steps captured using TestCase Studio
- vii. Website audit reports generated using Google Lighthouse for selected quality checks such as performance, accessibility, best practices, and SEO

The test items include the major functional modules of the system, namely:

- User Authentication Module
 - Patient registration
 - Patient login
 - Doctor login
 - Admin login

- Logout
- Patient Module
 - Book appointment
 - View appointment history
 - Cancel appointment
 - View prescription
 - Bill PDF generation
 - Submit contact query
- Doctor Module
 - View appointment
 - Prescribe medication
 - Cancel appointment
 - View prescription list
- Admin Module
 - Add doctor
 - Delete doctor
 - View and search doctor list
 - View and search patient list
 - View and search appointment details
 - View prescription list
 - View and search user queries

These test items will be used to verify that the implemented HMS features behave in accordance with the specified functional requirements, interface requirements, and

expected user workflows described in the SRS. Testing will include both manual validation and tool-supported testing activities, where Katalon Recorder supports browser-based automated workflow testing, TestCase Studio supports capture and documentation of test steps, and Google Lighthouse supports automated quality auditing of selected web pages.

2.2 Test Traceability Matrix

Table 2.1 maps the major HMS functional features to their corresponding module, estimated risk level, planned automated testing support, and responsible user role.

Feature ID	Risk Level	Module	Feature Description	Automated Testing Tool	Roles
F001	High	Authentication	Register Account	Katalon Recorder, TestCase Studio	Patient
F002	High	Authentication	Login	Katalon Recorder, TestCase Studio, Google Lighthouse	Patient, Doctor, Admin
F003	Medium	Authentication	Logout	Katalon Recorder, TestCase Studio	Patient, Doctor, Admin
F004	High	Patient Module	Book Appointment	Katalon Recorder, TestCase Studio	Patient

F005	Medium	Patient Module	View Appointment History	Katalon Recorder, TestCase Studio	Patient
F006	High	Patient Module	Cancel Appointment	Katalon Recorder, TestCase Studio	Patient
F007	High	Patient Module	View Prescription and Generate Bill PDF	Katalon Recorder, TestCase Studio	Patient
F008	Medium	Patient Module	Submit Contact Query	Katalon Recorder, TestCase Studio,	Patient
F009	Medium	Doctor Module	View Appointment	Katalon Recorder, TestCase Studio	Doctor
F010	High	Doctor Module	Prescribe Medication	Katalon Recorder, TestCase Studio	Doctor
F011	High	Doctor Module	Cancel Appointment	Katalon Recorder, TestCase Studio	Doctor
F012	Medium	Doctor Module	View Prescription List	Katalon Recorder,	Doctor

				TestCase Studio	
F013	High	Admin Module	Add Doctor	Katalon Recorder, TestCase Studio	Admin
F014	High	Admin Module	Delete Doctor	Katalon Recorder, TestCase Studio	Admin
F015	Medium	Admin Module	View & Search Doctor List	Katalon Recorder, TestCase Studio	Admin
F016	Medium	Admin Module	View & Search Patient List	Katalon Recorder, TestCase Studio	Admin
F017	High	Admin Module	View & Search Appointment Details	Katalon Recorder, TestCase Studio	Admin
F018	Medium	Admin Module	View Prescription List	Katalon Recorder, TestCase Studio	Admin
F019	Medium	Admin Module	View & Search User Queries	Katalon Recorder, TestCase Studio	Admin

Table 2.1 Test Traceability Matrix

2.3 Features to be Tested

Table 2.2 is based on the SRS and lists the features to be tested. The feature ID, feature description, and risk level are listed. The features are ranked according to risk based on whether these features are related to crucial workflows, data, role access, or operational processes.

Feature ID	Feature Description	Risk Level
F001	Register account functionality including input validation, password confirmation, and successful account creation for patients.	High
F002	Login functionality for patients, doctors, and admin, including credential validation, role-based access, and error messages.	High
F003	Logout functionality including session termination and redirection to the login page.	Medium
F004	Appointment booking functionality including specialization selection, doctor selection, fee display, date/time selection, and booking validation.	High
F005	Viewing appointment history including retrieval and display of doctor name, consultation fee, appointment date, time, and status.	Medium
F006	Patient appointment cancellation including confirmation prompt, status update, and correct cancellation handling.	High
F007	View prescription details, bill payment handling, and bill PDF generation for patient records.	High
F008	Contact query submission including form validation and successful message storage.	Medium
F009	Doctor appointment viewing including retrieval of assigned appointments and search by patient contact number.	Medium

F010	Prescription creation by doctor including entry and submission of disease, allergy, and prescription details for an active appointment.	High
F011	Doctor appointment cancellation including cancellation of active appointments and correct update of appointment status.	High
F012	Doctor prescription list viewing including retrieval and display of prescription records related to the doctor's patients.	Medium
F013	Add doctor functionality including validation and creation of a new doctor account by the admin.	High
F014	Delete doctor functionality including deletion of an existing doctor record and confirmation of handling.	High
F015	View and search doctor list functionality including retrieval of all doctor records and search by email.	Medium
F016	View and search patient list functionality including retrieval of all patient records and search by contact number.	Medium
F017	View and search for appointment details including retrieval of appointment records and filtering by patient contact number.	High
F018	Admin prescription list viewing including retrieval and display of prescription records for administrative review.	Medium
F019	View and search user queries including retrieval of submitted messages and filtering by contact number.	Medium

Table 2.2 Features to Be Tested

2.4 Features Not to Be Tested

This test plan does not cover the following aspects unless otherwise stated:

- i. Security testing
- ii. Cross-device responsiveness testing for all screen sizes and device categories

- iii. Third-party library reliability outside the application scope
- iv. Hardware-specific or medical device integration testing
- v. Non-functional attributes like reliability, performance, load testing

2.5 Approach

System testing for Hospital Management System (HMS) is mainly based on black-box testing. This strategy is used to test the functional behavior of how the system works as a black box according to the requirements, inputs, outputs and outcomes, without reference to its internal source code. The goal is to ensure that the system works as expected from an end user point of view.

In this test plan, risk-based testing will be used. Modules that are more important to the business and are more likely to have failure will be tested first. These will mainly include the modules of real-time identity verification, request an appointment, cancel an appointment, manage prescription, manage doctors, and manage appointment records. Features with medium risk, such as viewing history, submitting message and searching, will also be tested after testing the primary functions.

The black-box testing techniques used in this test plan include:

- i. **Equivalence Partitioning**
Used to divide input data into valid and invalid groups, such as registration details, login credentials, contact numbers, and search values.
- ii. **Boundary Value Analysis**
Used for fields and conditions with meaningful boundaries, such as password length, required fields, and date-related inputs.
- iii. **State Transition Testing**
Used for workflows involving state changes, such as login/logout, appointment booking, appointment cancellation, and status updates.

iv. **Decision Table Testing**

Used for functions that depend on combinations of conditions and actions, such as login validation, appointment validation, and role-based access behavior.

v. **Use Case Testing**

Used to validate end-to-end functional workflows based on the use cases identified in the Software Requirements Specification (SRS), including patient, doctor, and admin operations.

vi. **Error Guessing**

Used to identify likely fault scenarios based on tester experience, such as empty fields, invalid credentials, duplicate submissions, and invalid search input.

Testing will include both manual testing and tool-supported testing activities. Manual testing will be used to validate complete workflows, abnormal scenarios, user-visible outputs, and system behavior that requires human observation. Tool-supported testing will be used to improve efficiency, repeatability, and evidence collection during the testing process.

The selected tools will be applied as follows:

- **Katalon Recorder** will be used to record and replay browser-based functional workflows such as registration, login, appointment booking, cancellation, and admin operations.
- **TestCase Studio** will be used to capture test steps, screenshots, and detailed action records to support test case documentation and execution evidence.
- **Google Lighthouse** will be used to perform automated audits on selected web pages for quality attributes such as performance, accessibility, best practices, and SEO.

The testing activities will be carried out in the following phases:

Phase 1: Authentication and Basic Access Testing

- Test patient registration, login, and logout.

- Test doctor login and admin login.
- Verify role-based access and input validation.

Phase 2: Core Functional Module Testing

- Test the patient, doctor, and admin modules individually.
- Verify appointment booking, appointment cancellation, prescription handling, doctor management, and record viewing/searching.

Phase 3: Integrated Workflow and Quality Checking

- Test integrated workflows across modules, such as patient appointment booking followed by doctor appointment viewing and prescription creation.
- Validate patient prescription viewing and bill generation.
- Perform selected website quality audits using Google Lighthouse.

All testing activities will be conducted in a controlled test environment that closely resembles the intended HMS operating environment. This is to ensure that the testing results are consistent, repeatable, and relevant to the actual system behavior.

2.6 Item Pass / Fail Criteria

The system will be considered as passed if it satisfied the following criteria:

- i. Each executed test case shall produce the expected result as defined in the test case or related requirement.
- ii. All mandatory functionalities shall behave in accordance with the Software Requirements Specification (SRS).
- iii. Required input fields, validations, and workflows shall function correctly for both valid and invalid conditions.
- iv. Appropriate error messages, warning messages, or confirmation messages shall be displayed whenever necessary.

- v. No system crash, fatal execution failure, or unrecoverable application error shall occur during normal or abnormal user operations.
- vi. Core modules such as authentication, appointment handling, prescription management, and admin operations shall not contain unresolved critical defects.
- vii. For tool-supported testing, the recorded or captured execution results from Katalon Recorder, TestCase Studio, and Google Lighthouse shall be consistent with the observed system behavior and expected test outcome.

The system will be considered as failed if it meets the following criteria:

- i. The actual system behavior does not match the expected result.
- ii. The required function cannot be completed successfully.
- iii. Validation rules are not enforced correctly.
- iv. Incorrect or missing system messages are observed.
- v. The system produces incorrect data, inconsistent output, or an application error during execution.
- vi. A critical or major defect prevents completion of the intended workflow.

2.7 Suspension Criteria and Resumption Requirements

2.7.1 Suspension Criteria

- i. A critical system failure causes the HMS application to become unavailable.
- ii. A database connection failure, server-side error, routing error, or configuration issue prevents further test execution.
- iii. The test environment becomes unstable, inaccessible, or inconsistent with the expected setup.
- iv. A blocking defect in a major module prevents progression to subsequent test cases or workflows.

- v. Required test accounts, test data, or necessary system pages are no longer accessible.
- vi. Tool-supported testing cannot proceed because Katalon Recorder, TestCase Studio, or Google Lighthouse is unavailable or not functioning correctly for the intended test activity.
- vii. A defect causes repeated crashes, corrupted results, or invalid execution evidence, making further testing unreliable.

2.7.2 Resumption Requirements

- i. The HMS application has been restored to a stable and testable state.
- ii. The database, application routes, interfaces, and required services are accessible again.
- iii. Blocking defects have been fixed, or an acceptable workaround has been provided and documented.
- iv. The test environment has been revalidated before resuming execution.
- v. Required test accounts, test data, and supporting modules are available again.
- vi. The selected testing tools, namely Katalon Recorder, TestCase Studio, and Google Lighthouse, are functioning properly for the relevant testing tasks.

2.8 Test Deliverables

Test documentation:

- i. Test Plan
- ii. Test Design Specification
- iii. Propose automated testing / Test management tool

2.9 Entry Criteria

Testing may begin only when the following conditions are satisfied:

- i. The approved Software Requirements Specification (SRS) is available.
- ii. The HMS application build is available and accessible for testing.
- iii. The test environment is configured and functioning properly.
- iv. Required testing tools, including Katalon Recorder, TestCase Studio, and Google Lighthouse, are installed and ready to use.
- v. Test cases and required test data are prepared.

2.10 Exit Criteria

Testing may be concluded only when the following conditions are satisfied:

- i. All planned test cases have been executed.
- ii. Test results and supporting evidence have been recorded.
- iii. Critical defects have been fixed, retested, or formally accepted.
- iv. Core HMS functions comply with the acceptance criteria defined in the SRS.
- v. Test deliverables are completed and ready for review.

3.0 Test Management

3.1 Planned Activities and Tasks

Table 3.1 shows the necessary tasks for preparing and performing testing for the Hospital Management System (HMS).

No	Task	Activities	Target Date
1	Test Planning and Control	• Define test scope • Review SRS and system modules • Identify risks • Confirm selected testing tools • Prepare test plan, conduct review, and obtain approval	20/4/2026
2	Test Design	• Identify test conditions • Design manual and black-box test cases • Prepare test data • Map test cases to the HMS functional requirements	22/4/2026
3	Test Environment Setup	• Configure the HMS test environment • Prepare browser and local server access • Verify test accounts • Set up automated testing tools	24/4/2026
4	Test Execution	• Execute manual and tool-supported test cases for authentication, patient, doctor, and admin modules • Capture results and collect execution evidence.	25/4/2026

5	Defect Reporting	- Record detected defects - Attach screenshots or supporting evidence - Classify defect severity, communicate findings, and retest corrected issues.	27/4/2026
6	Test Closure	- Review test results - Confirm completion of planned testing activities - Finalize the test summary report	29/4/2026

Table 3.1 Planned Activities & Tasks

3.2 Environment & Infrastructure

This section outlines the environment and supporting infrastructure required to test the Hospital Management System (HMS) effectively.

3.2.1 Hardware Requirements

The following hardware components are required for the test environment:

- Processor: Intel Core i5 / AMD Ryzen 5 or equivalent
- RAM: Minimum 8 GB
- Storage: At least 20 GB available space
- Display: 1366 x 768 resolution or higher
- Network: Stable internet connection for testing

3.2.2 Software

The following software is required for testing HMS:

- Operating System: Windows 10 or Windows 11

- Web Server: Apache via XAMPP
- Programming Environment: PHP
- Database: MySQL / MariaDB
- Browser: Google chrome
- IDE / Editor: Visual Studio Code
- Testing Tools: Katalon Recorder, TestCase Studio, Google Lighthouse

3.2.3 Publications

The following documents and materials are required during testing:

- Software Requirements Specification (SRS)
 - Outlines the functional and non-functional requirements of the system.
- Test Plan
 - Defines the approach and schedule of testing activities.
- Test Design Specifications
 - Further refine test approach from Test Plan, define the sources of the information used to prepare the plan and identify the general cases of test to be executed.

3.3 Risk & Contingency

This section identifies the risk issues that may impact successful completion of the system testing activities of the Hospital Management System (HMS). The potential likelihood, impact and mitigation plan were described in the following tables:

Likelihood Rating:	
1	Low

2	Medium
3	High
4	Extreme

Impact Rating:	
1	Low
2	Medium
3	High
4	Extreme

Functiona l ID	Risk Description	Likelihoo d	Impac t	Risk Exposur e	Risk Treatment Plan	People Responsibl e
F002	User unable to log in with valid credentials	4	4	16	Verify login inputs, credential validation, role-based access, and error handling through manual and tool-supported testing	PANG TIAN YOU
F004	Appointmen t booking fails or stores	4	4	16	Test valid and invalid booking flows,	PANG TIAN YOU

	incorrect appointment details				date/time validation, doctor selection, and confirmation behavior	
F010	Prescription details are not saved correctly	3	4	12	Test prescription entry with valid and invalid inputs, confirm database update, and verify displayed prescription records	PANG TIAN YOU
F013	Admin unable to add a doctor or doctor record stored incorrectly	3	4	12	Test doctor creation with valid and invalid data, check field validation, and verify successful record creation	PANG TIAN YOU

F014	Admin deletes wrong doctor record or deletes function fails	3	4	12	Verify deletion workflow, confirmation handling, and correct record removal based on existing doctor data	PANG TIAN YOU
F017	Appointment details cannot be viewed or searched correctly by admin	3	3	9	Test full appointment list retrieval, search by contact number, and validation of displayed appointment status	PANG TIAN YOU
F007	Bill PDF cannot be generated after prescription handling	2	4	8	Test bill generation flow after valid prescription and ensure output is produced	PANG TIAN YOU

					correctly in PDF format	
F019	Submitted user queries cannot be retrieved or searched by admin	2	3	6	Test query list retrieval, search input handling, and display of matching query records	PANG TIAN YOU

4.0 General

4.1 Metrics

The following metrics will be collected and monitored throughout the testing process to evaluate progress, quality, and coverage of the Hospital Management System (HMS).

Metric List	Metric Description	Goals
Actual duration vs plan	Compares actual testing progress against the planned schedule	Within planned schedule or acceptable delay
Percentage of test cases executed	Measures how many planned test cases have been executed	100%
Percentage of test cases passed	Measures how many executed test cases passed successfully	All high-risk test cases passed
Percentage of requirements covered	Measures how many HMS requirements are covered by test cases	100% coverage
Defect count by severity	Tracks the number of defects found based on severity level	As low as possible
Defect closure rate	Measures how many reported defects have been fixed and closed	100% closure for critical defects

4.2 Glossary

Term	Definition
SRS	Software Requirements Specification
TP	Test Plan
HMS	Hospital Management System

4.3 Incident Classification

Impact	Mission Critical: The system fails to operate, or core HMS functions such as login, appointment booking, prescription handling, or admin operations cannot be completed.
	Major: A serious defect affects important functionality, but a workaround may still exist.
	Minor: A defect exists but does not significantly affect the main operation of the system.
	Cosmetic: A visual or layout issue that does not affect system functionality.
Priority	Immediate: Must be fixed immediately because it blocks testing or critical system operation.
	High: Must be fixed before the next testing phase or submission.
	Medium: Should be fixed soon, but testing may continue with caution.
	Deferred: May be postponed if time or project constraints do not allow immediate correction.